INSTRUCTIONS: Answer ALL of the following questions.

Question 1 **(15 marks)**

a) Requirement volatility is one of the major software engineering design challenges. **Describe** the **Requirement Volatility** challenge and **Identify** another **TWO(2)** major software engineering design challenges with its descriptions.

(5 Marks)

b) A software design process involves a series of actions and regulations that define how resources collaborate to create software design artifacts. **Describe TWO(2)** major activities of the software design process.

(5 Marks)

c) Software design principles establish the foundation for creating reliable, maintainable, and scalable software systems that meet the needs of users and stakeholders while being adaptable to changing requirements and technological landscapes. One of the design principles is Abstraction. **Explain** how the **Abstraction** is conducted and **Identify** another **ONE(1)** software design principle based on your learning with its detailed explanation.

(5 Marks)

Question 2 **(25 marks)**

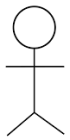
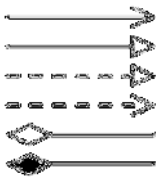
a) **Define** what is a Unified Modelling Language (UML).

(2 Marks)

b) **Identify THREE(3)** basic components used in UML diagrams as shown in **Table 1**.

(3 Marks)

Table 1 UML Basic Components

No	UML Components	Descriptions
1		
2		
3		

c) Given the class diagram description in **Table 2** for a Library Management System, **Design** the complete Class Diagram.

(10 Marks)

Table 2 Class Diagrams Descriptions

Class Name	Attributes	Methods	Description
Book	- bookID - title - status	+ checkAvailability() + updateStatus()	Represents the book, including ID, title, and status (available/checked out).
Patron	- patronID - name - accountStatus	+ requestBook() + checkEligibility()	Represents the library user who requests to borrow books.
Librarian	- librarianID - name	+ validatePatron() + recordTransaction()	Represents the librarian who facilitates the borrowing process.
LibrarySystem	- transactionID - dueDate	+ saveTransaction() + generateDueDate()	Manages transactions and assigns due dates for borrowed books.
Transaction	- transactionID - borrowDate - dueDate	+ createTransaction()	Represents each borrowing transaction, including dates and associated book and patron.

d) Given the "Borrow Book" use case description in **Table 3** for a Library Management System, **Design** a complete Sequence Diagram.

(10 Marks)

Table 3 Borrow Book Use Case Specifications.

Attribute	Details
Use Case Name	Borrow Book
Description	A library system allows a user (patron) to borrow a book, recording the transaction and updating book availability.
Actors	Patron, Librarian, Library System
Preconditions	1. Patron has a valid library account. 2. Book is available in the library for borrowing.
Basic Flow	1. Patron requests to borrow a book. 2. Librarian checks book availability. 3. System validates patron eligibility. 4. Librarian records the borrowing transaction. 5. System updates book status to "checked out". 6. System provides a due date for return.
Alternate Paths	If the book is not available, the system allows the patron to reserve the book.
Postconditions	1. The book's status is updated to "checked out". 2. Transaction details are saved in the library system.
Extension Points	<Extend Loan Duration> <Return Book>

Question 3 **(20 marks)**

a) **Identify** and **Explain TWO(2**) main tasks in the process of developing software architectures for a project.

(5 Marks)

b) **Identify** and **Describe TWO(2**) quality in software design and architecture.

(5 Marks)

c) Consider the project in **Question 2, (c) and (d),** the Library Management System, apply the concept of 4+1 architectural views to **Explain** how different **Views** (logical, development, process, physical, and user) can be utilized to address specific challenges encountered during the project's development. **Discuss** the **advantages** of employing these views.

(10 Marks)

SECTION B - PROBLEM SOLVING

INSTRUCTIONS: Answer ALL the following questions. **(40 marks)**

Question 4 **(25 marks)**

Project 1: E-Commerce Platform

A large e-commerce platform needs an architecture that can handle a massive volume of transactions, support various product categories, and allow different teams to develop features independently. It should be resilient, so failures in one part of the system don't affect the entire application. This platform should be able to adapt to new data inputs quickly to keep recommendations relevant and updated.

Project 2: University Student Portal

A university wants to develop a student portal that serves as a central hub for students to access their academic records, assignments, course materials, and university announcements. The portal must support various user roles (students, faculty, administrators) with personalized dashboards and secure access to resources based on each role. The portal should also facilitate interaction through features like announcements, discussion boards, and assignment submissions.

a) **Analyze** the requirements and **identify** a suitable architectural style for each project above.

(4 marks)

b) **Evaluate** for both chosen architecture styles in **(a). Discuss** how the architectural styles would address the requirements.

(4 marks)

c) **Choose ONE(1)** project and **Design** the class diagram for the architecture style. The class diagram must reflect your architecture style answer in **(a).**

(9 marks)

d) Based on the class diagram answered in **(c)**, design and write the code representing the class diagram.

(8 marks)

Question 5 **(15 marks)**

Project 1: E-Commerce Platform

Extension for the E-Commerce Platform

The e-commerce platform needs a modular recommendation engine that can quickly adapt to new product categories and custom recommendation algorithms. The recommendation engine should allow future developers to add new algorithms or product types without modifying the core engine code.

Project 2: University Student Portal

Extension for the University Student Portal

The university portal should have a feature where students can access course materials, but each course type has different access needs. For example, online courses might include interactive quizzes, while in-person courses may link to downloadable materials. The system should allow flexibility in handling each course type while maintaining consistent functionality across all courses.

a) **Analyze** the extensions for both projects and suggest a suitable design principle for internal component design for each project.

(4 marks)

b) **Evaluate** for both chosen design principles in **(a). Discuss** how the application of the design principle would benefit the project.

(4 marks)

c) Choose 1 design principle in **(a). Design** the class diagram to illustrate the design principle application.

(7 marks)

- QUESTIONS END -